Md Abdullah Hossien Molla

B.Sc. in Computer Science | Prospective Graduate Researcher
Independent University, Bangladesh
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Research Profile

Final-year Computer Science student at Independent University, Bangladesh with hands-on research expertise in Computer Vision, Deep Learning, and Applied Machine Learning. Author of 4 peer-reviewed publications across IEEE and Springer venues (3 as First Author) spanning object detection, clinical ML, and assistive robotics. Experienced across the full research lifecycle from dataset curation and PyTorch/YOLO benchmarking to peer-reviewed manuscript publication. Seeking a research-oriented Master's program to contribute to laboratory research in visual computing, intelligent systems, and deployable AI.

Research Interests

Computer Vision and Scene Understanding; Object Detection and Spatial Feature Representation; Applied Deep Learning for Assistive and Healthcare Systems; Uncertainty-Aware and Safety-Critical ML; Data Governance and Reproducibility in AI Pipelines. *Current focus: benchmark-driven object detection, real-time edge vision pipelines, and deployable AI systems for real-world applications.*

Education

- **Independent University, Bangladesh** — *2022 – 2026 (Expected)*
  Bachelor of Science in Computer Science (CGPA: 3.64 / 4.00, 128 Credits Earned) — Dhaka, Bangladesh
- **Govt. Science College, Tejgaon** — *2020*
  Higher Secondary Certificate (HSC) (GPA: 5.00 / 5.00) — Dhaka, Bangladesh

Publications

- ***Peer-Reviewed Conference Papers***
- **Road Condition Detection on Bangladeshi Roads: A Benchmark of YOLO Architectures**
  ICSIE 2026 | *Accepted, in press* | *First Author*

  Benchmarked YOLOv5s/v8s/v10s/v11s across 11,750 road-defect instances (4,254 augmented images, 6 classes); binary formulation achieved 71.1% mAP50 while reducing training time by 23%. YOLOv11s achieved optimal accuracy at 9.5 ms inference and a 19.2 MB footprint on a Tesla T4 GPU, demonstrating that scaling resolution from 640px to 768px doubled training time without accuracy gain.

- **Braille for All: A $70 AI-Powered Braille Tutoring Robot That Erases the Literacy Gap with a Touch and a Voice**
  IEEE CCE 2026, Mexico City | *DOI: 10.1109/CCE72160.2026.11734982* | *First Author*

  Engineered an AI-powered tutor robot with ISO-compliant refreshable cells (48 ms actuation, 0.72 mm protrusion, 4.8 N force) and on-device Bangla ASR (94.2% accuracy on 39 consonants); a 10-week longitudinal trial with 87 blind students showed 2.83× faster letter recognition with a System Usability Score of 82.4/100.

- **Safety-Enhanced Machine Learning for Fetal Health Classification Using Cardiotocography Data**
  IEEE ICMLAS 2026 | *DOI: 10.1109/ICMLAS67792.2026.11483766* | *First Author*

  Benchmarked five classifiers across 2,126 CTG records; XGBoost achieved 94.13% accuracy (98% Normal, 91% Pathological recall). Designed a confidence-based safety gateway with a 0.7-threshold to flag uncertain predictions for clinical review, significantly cutting false negatives.

- ***Book Chapter***
- **Understanding Challenges of Liver Data Governance for Mobile App-Based Liver Care in Bangladesh**
  Springer LNCS, HCII 2026, pp. 120–138 | *DOI: 10.1007/978-3-032-30549-7_7* | *Co-Author*

  Resulted from Sponsored Research Project 2025-SETS:R15. Combined a PRISMA systematic review with stakeholder interviews across Dhaka to map fragmented clinical data flows, proposing a UTAUT–governance hybrid framework for the National Hepatology Data System (NHDS) pipeline.

Research & Academic Appointments

- **Undergraduate Research Assistant** *Aug 17, 2025 – May 15, 2026*
  Sponsored Research Project: 2025-SETS:R15 Independent University, Bangladesh
  - Appointed for two consecutive terms (20 hrs/week on-site, 80 hrs/month) to conduct technical problem formulation, literature synthesis, and qualitative data gathering.
  - Executed semi-structured stakeholder interviews and qualitative analyses that formed the empirical basis for our Springer LNCS (HCII 2026) publication.
  - Maintained strict compliance with institutional research integrity, data ethics protocols, and project milestone deliverables.

- **Undergraduate Teaching Assistant / Student on Duty (SOD)** *Autumn 2024 – Autumn 2025*
  Dept. of Computer Science and Engineering — CIS 101 / CSC 101 Independent University, Bangladesh
  - Delivered theory tutorial sessions, review hours, and conceptual problem-solving hours across four consecutive semesters (Autumn 2024, Spring 2025, Summer 2025, Autumn 2025).
  - Mentored hundreds of students in foundational Python programming paradigms, algorithmic problem breakdown, data structures, and modular function design.
  - Proctored major examinations, graded theory problem sets, and tabulated continuous assessment scores.

Selected Research & Technical Projects

- **Road Condition Detection & Scene Parsing System** *2025*
  Computer Vision Python, PyTorch, OpenCV, CUDA
  - Engineered an end-to-end vision training and inference harness benchmarking YOLOv5s, v8s, v10s, and v11s detector backbones across complex road topologies.
  - Conducted systematic multi-scale feature anchor evaluations, hyperparameter ablation studies, and spatial confusion matrix analyses for road anomaly segmentation.

- **GloHealth AI — Interpretable Machine Learning Platform** *2025*
  Applied ML Python, Scikit-learn, Flask, Supabase
  - Implemented inference endpoints backed by Scikit-learn and XGBoost, outputting predictive probabilities and feature importance metrics.
  - Architected relational schema on PostgreSQL/Supabase and containerized client-server interfaces on cloud infrastructure.

Technical Skills

Languages: Python, C, C++, SQL, JavaScript.
Computer Vision & ML: PyTorch, Ultralytics YOLO, OpenCV, Scikit-learn, Torchvision, CUDA, XGBoost.
Hardware & Systems: Digital Logic Design, Microcontrollers, Embedded Interfacing, System Architecture, Linux/Bash.
Mathematics & Computing: Linear Algebra, Coordinate Transformations, Multivariable Calculus, Numerical Methods, ODEs.
Developer Tools: Git/GitHub, Docker, LaTeX, Jupyter, Google Colab, Supabase, VS Code.
Core Research Competencies: Empirical benchmark design, failure mode analysis, hardware-software integration, academic writing.

Honors & Academic Achievements

- Merit-based Undergraduate Tuition Scholarship, Independent University, Bangladesh.
- Dean's Merit List Award for sustained academic excellence in Computer Science.
- Official FIDE Rapid Chess Rating holder; Active competitive tournament player.
- Event Coordinator, July Memorial FIDE Chess Championship 2025 — managed logistics and tournament scheduling.

Academic References

- **Md Tarek Habib, PhD**
  Associate Professor, Dept. of Computer Science and Engineering, School of Engineering, Technology and Sciences
  Independent University, Bangladesh
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- **Mohammad Shidujaman, PhD**
  Assistant Professor, Dept. of Computer Science and Engineering, School of Engineering, Technology and Sciences
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